



Department of Electronics and Communication Engineering

Academic Year 2021 – 2022 (Odd Semester)

Degree, Semester & Branch: B.E., V & ECE B

Course Code & Title: EC8552 & Computer Architecture and Organization

Name of the Faculty member (s): Dr.A.Azhagu Jaisudhan Pazhani

Innovative Practice Description

- **Unit / Topic:** I/ Splitting and reconstructing central processing unit
- **Course Outcome:** CO1
- **Topic Learning Outcome:** TLO1
- **Activity Chosen:** Hardware Demonstration
- **Justification:**

A method demonstration is a teaching approach that uses visual aids such as flip charts, posters, and power point to express an idea. A demonstration is a step-by-step technique for showing someone how to manufacture or execute something.

Demonstration as a teaching method benefits students by providing a better learning experience in the classroom, generating interest in the subject, assisting in the development of the spirit of inquiry, students cooperating in the teaching-learning process, and the learning process becoming permanent in the student's life.

Time Allotted for the Activity: 30 minutes

- **Details of the Implementation:**

The computer system's hardware includes the monitor, CPU, keyboard, mouse, printer, sound system, RAM, hard disc, and other components. Hardware is used to take data from a user, process it, and show the results, as well as to execute orders given by an individual. The computer may be put together in the lab, and the internal components of the computer system can be discussed in detail. Students were given reference books and DVDs as well as other materials.

CO – PO / PSO mapping:

CO	PO1	PO2	PO3	PO4	PO10	PO11	PO12	PSO3
CO1	3	3	2	2	2	2	1	1

- **PO / PSO mapped:**

Innovative practice	PO1	PSO3
	3	1
Justification for correlation	Students must have a basic understanding of engineering basics in order to comprehend the computer's functional unit. As a result, it's assigned to level 3.	Students must have a basic understanding of engineering basics in order to comprehend the computer's functional unit. As a result, it's assigned to level 3.

- **Images / Screenshot of the practice:**



- **Reflective Critique:**

- ❖ ***Feedback of practice from students and other stakeholders:***

Students believe that their self-learning has improved. They learn how to communicate with the computer's processing unit.

- ❖ ***Benefit of the practice:***

During the session, the majority of the students were perplexed by the internal components of the computer system. Following the demonstration, they will have a better understanding of the computer system's processing unit. Some students were uninterested throughout the demonstration, so I counselled them individually and explained the necessity of hardware demonstrations.

- ❖ ***Challenges faced in implementation:***

I had planned for the activity to last 20 minutes, but some of the students had questions about the internal workings of the computer system, which I was able to answer for another 10 minutes.

References:

- ❖ David A. Patterson and John L. Hennessey, "Computer Organization and Design", Fifth edition, Morgan Kauffman/Elsevier, 2014.
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Innovative Practice Description

- **Unit / Topic:** V/ Multiprocessors Network Topologies
- **Course Outcome:** CO5
- **Topic Learning Outcome:** TLO16
- **Activity Chosen:** Laboratory Visit (NS-2 Simulation)
- **Justification:**

NS-2 is used to correlate theoretical concepts utilising a simulation programme. It allows students to witness the tool as it is being utilised in the Communication Networks laboratory during a later semester. As a result, they must be aware of it.

- **Time Allotted for the Activity:** 30 minutes
- **Details of the Implementation:**

First, I went through the foundations of the NS-2 simulator, the different sorts of topologies, and how to create basic Ns-2 coding. I input the coding for bus topology and executed the output using that simulation software.

Simulate the Ring and Star topologies in the same way, and present the results. This job will take longer to accomplish since you will need to grasp the step-by-step approach to acquire the desired outcome. As a result, you will not be able to complete it in the time allotted because of the time it will take to give the procedures.

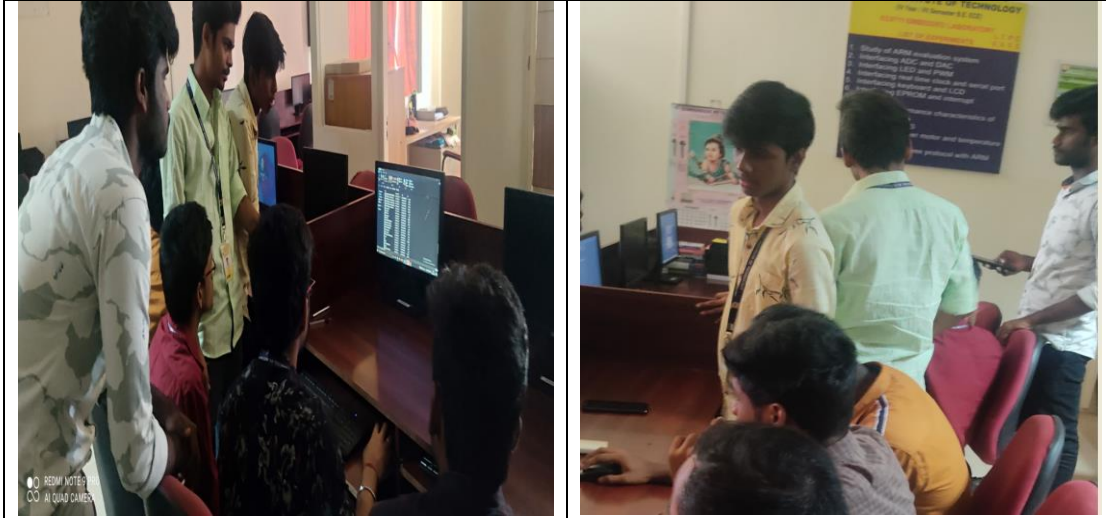
CO – PO / PSO mapping:

CO	PO1	PO2	PO3	PO4	PO10	PO11	PO12	PSO3
CO5	3	3	2	3	2	2	1	1

- **PO / PSO mapped:**

Innovative practice	PO11	PSO3
	2	1
Justification for correlation	Students will be able to use NS-2 to demonstrate their understanding of multiprocessor network architectures. As a result, it's mapped to level 2.	Students will be able to construct and analyse many types of multiprocessor network topologies. As a result, it's mapped at level 1.

- **Images / Screenshot of the practice:**



- **Reflective Critique:**

- ❖ ***Feedback of practice from students and other stakeholders:***

Students believe that their self-learning has improved. They will learn how to implement the various topologies that are used in Multiprocessor Configuration.

- ❖ ***Benefit of the practice:***

During class, the majority of the pupils were perplexed by the notion of topology. They can gain a better understanding of the different sorts of Multiprocessor Network Topologies after a laboratory tour with demonstration. Some students were not engaged during the demonstration, so I counselled them individually and explained the necessity of topologies in Advanced Computer Architecture. Students may be able to analyse many sorts of layout for various topologies with this tool.

- ❖ ***Challenges faced in implementation:***

I had planned for the session to last 30 minutes, but several students had questions about NS-2 basics and structural flow of code, so I clarified it for another 10 minutes. Because the students have trouble following the technique, it takes longer to teach the concepts.

References:

- ❖ Miles J. Murdocca and Vincent P. Heuring, “Computer Architecture and Organization: An Integrated approach”, Second edition, Wiley India Pvt Ltd, 2015
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Academic Year: 2021- 2022 (Odd Semester)

INNOVATIVE TEACHING METHOD

Degree, Semester& Branch: V Semester B.E.ECE B

Course Code & Title: EC8552 Computer Architecture and Organization

Name of the Faculty member: Dr.A.Azhagu Jaisudhan Pazhani

Innovative Teaching Method Execution

UNIT I – COMPUTER ORGANIZATION AND INSTRUCTIONS

Computer System Performance – One Minute Paper on 27.08.2021



Addressing Modes – Mind Map on 03.09.2021



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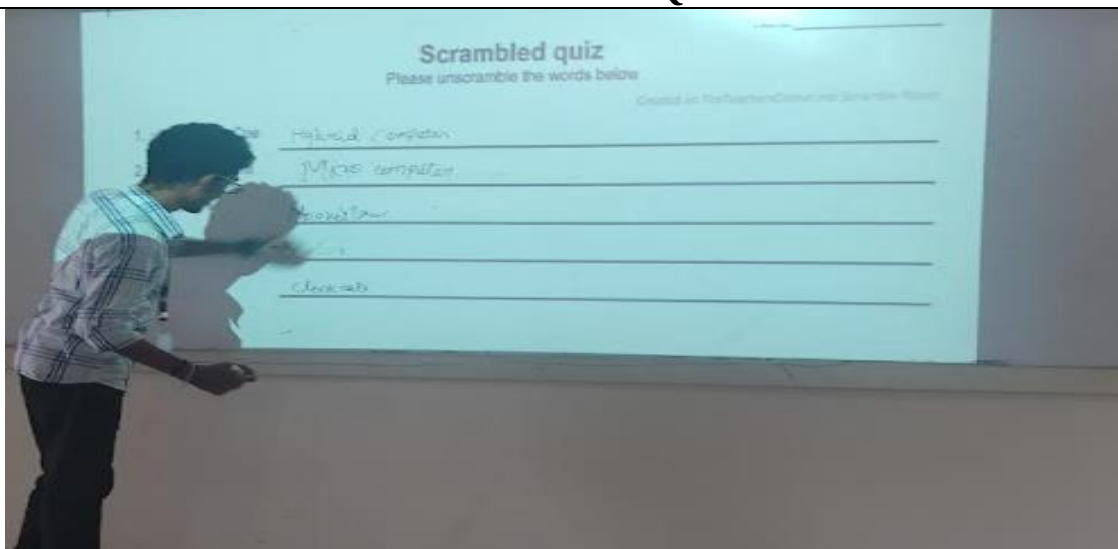
Innovative Teaching Method Execution

UNIT II – COMPUTER ORGANIZATION AND INSTRUCTIONS

Floating point arithmetic-open book test on 24.09.2021



Subword Parallelism – Visual Quiz on 04.10.2021



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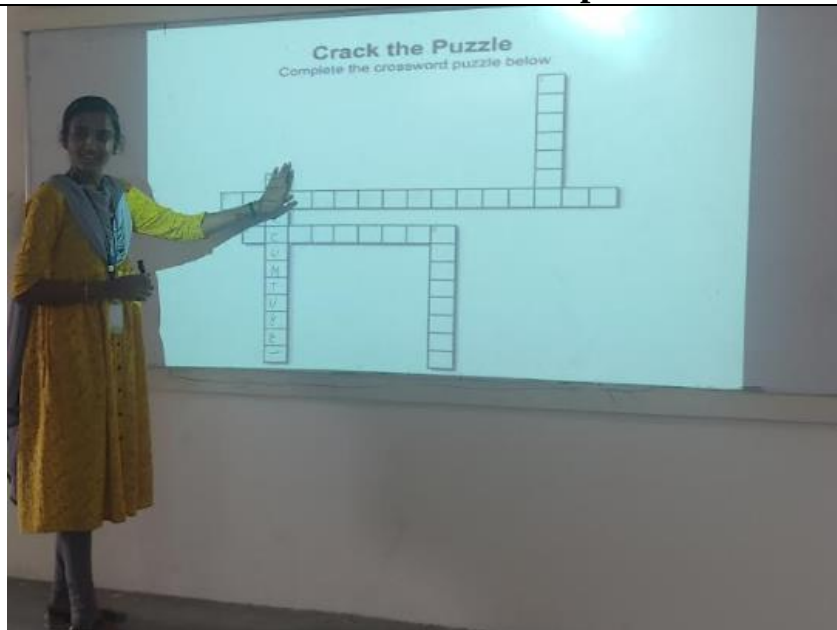
Innovative Teaching Method Execution

UNIT III – THE PROCESSOR

Logic Design Convention-concept test on 08.10.2021



Parallelism via instruction – crossword puzzle on 25.10.2021



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Innovative Teaching Method Execution

UNIT IV – MEMORY AND I/O ORGANIZATION

Internal Communication Methodology- class poll on 18.11.2021



I/O devices – Think Pair Share on 21.11.2021



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Innovative Teaching Method Execution

UNIT V – ADVANCED COMPUTER ARCHITECTURE

Parallel processing architecture - Flipped Classroom on 23.11.2021



Introduction to Graphic processing unit – Concept Test on 25.11.2021





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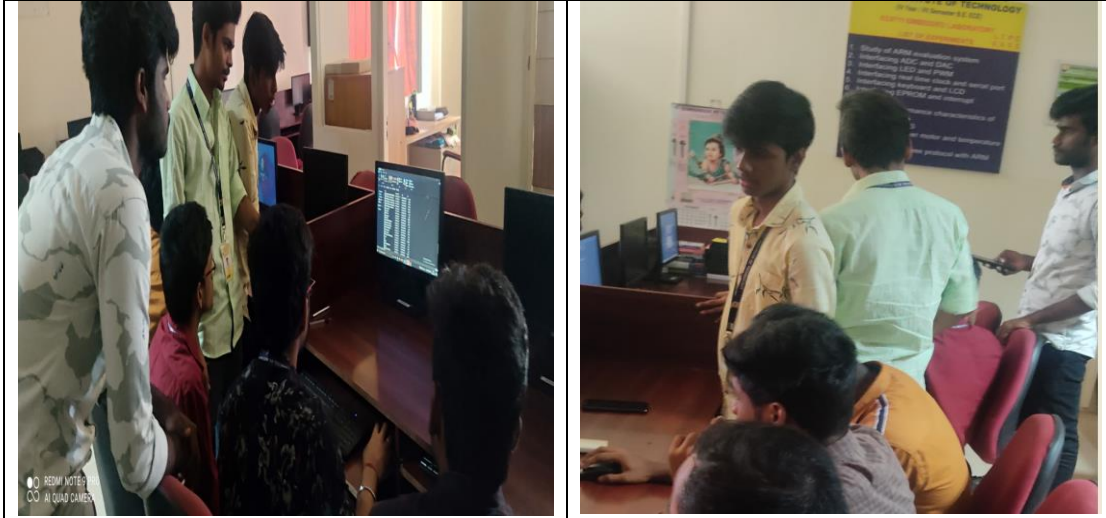
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